Nomination Citation – Societal Impact and Contribution Alumni Award

Biography:

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Tooba Ashiq Seyal is nominated for the COMSATS University Islamabad Alumni Excellence Award in *Societal Impact and Contribution* in recognition of her nearly decade-long academic association with COMSATS University Islamabad (2016–present) and her sustained, innovative, and socially impactful biomedical research addressing critical public health challenges in Pakistan and other developing regions. Across her BS, MS, and ongoing PhD training at COMSATS, she has consistently demonstrated academic excellence and a strong commitment to translating research into affordable, accessible, and societally relevant healthcare solutions. Notably, she chose COMSATS University Islamabad over offers from Quaid-i-Azam University and NUST, reflecting her deep dedication to advancing indigenous research and contributing to her alma mater's mission.

Ms. Seyal began her academic journey with a BS in Biosciences at COMSATS University Islamabad (2016–2020), where she gained a solid foundation in molecular biology and biomedical sciences. Her BS research on protein structure analysis of selected mutations in human Lipoprotein Lipase provided insights into rare SNPs, identifying deleterious

variants with potential disease associations, demonstrating her early focus on research with translational relevance for human health.

Building on this foundation, her MS research (2020–2022) represented one of the first interdisciplinary studies in Pakistan to investigate substance abuse using demographic surveys, molecular genetics, and in-silico analyses. Her thesis, *"In Silico and Molecular Analysis of GSTT1 and GSTM1 Genes in Drug-Addicted Cohorts in Pakistani Populations,"* revealed that young individuals from middle-income groups are particularly vulnerable to drug addiction. She identified GSTT1 polymorphisms as significantly more prevalent among addicted individuals, providing new molecular insights into genetic susceptibility, detoxification pathways, and addiction biology. This research has societal relevance in informing early risk identification, targeted prevention strategies, and public health policy, offering a scientific foundation for mitigating addiction in vulnerable populations.

Currently, Ms. Seyal is pursuing her PhD in Molecular Biology and Biochemistry at COMSATS University Islamabad (2022–ongoing). Her doctoral research focuses on designing pro-angiogenic biomaterials for wound healing—a pioneering effort in Pakistan addressing chronic ulcers, burn injuries, and infected wounds in resource-limited settings.

A particularly novel contribution of her PhD work is the **development of multifunctional electrospun polyurethane nanofibrous dressings** co-loaded with lidocaine and 2-deoxy-D-ribose, uniquely combining **analgesic and pro-angiogenic properties in a single dressing**—a system not previously reported. These dressings offer rapid pain relief and enhanced vascularization, demonstrating up to 98% wound closure in full-thickness rat models, with histological evidence of improved collagen deposition, re-epithelialization, and reduced inflammation. This work has the potential to **transform burn and chronic wound care in underserved communities**, reducing patient suffering and healthcare costs. A future patent is anticipated for this innovative dressing technology.

In parallel, she has developed a **dual-function sodium alginate–collagen hydrogel co-loaded with ciprofloxacin and 2-deoxy-D-ribose**, overcoming ciprofloxacin-induced cytotoxicity while retaining antimicrobial potency. This hydrogel effectively treats infected wounds caused by both Gram-positive and Gram-negative pathogens, including MRSA, while promoting cytocompatibility and angiogenesis. This novel platform has strong **translational and commercialization potential**, and a future patent is planned to secure intellectual property rights for this therapeutic innovation.

Collectively, Ms. Seyal's work exemplifies research with **direct societal impact**:

- **BS work** enhanced understanding of disease-linked SNPs, contributing to molecular diagnostics.

- **MS research** informed public health strategies for addiction prevention in Pakistan.
- **PhD work** directly addresses chronic and burn wounds in resource-limited communities through cost-effective, multifunctional biomedical products, offering tangible benefits for patients and healthcare systems.

Her research is distinguished by its **novelty, interdisciplinary integration of molecular genetics, bioinformatics, and biomaterials engineering**, and its clear societal relevance. Ms. Seyal's commitment to COMSATS University Islamabad, her focus on indigenous innovation, and her dedication to addressing critical healthcare challenges make her a highly deserving candidate for the **Alumni Excellence Award in Societal Impact and Contribution**.

References: Apart from 1 publication already in 4.5 Impact factor, : "Engineering lidocaine hydrochloride and 2-deoxy-D-ribose releasing electrospun nanofibrous dressings for pain relief and angiogenesis – Accelerated wound healing in rat mode" in distinguished Journal: "Materials Today Communications", recently published in 2025. Over 1 more publication are expected from her PhD work, with patent submissions in preparation for her multifunctional wound dressings and hydrogels.